

Bull and Bear Stock

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DATED: MAY 2025



Content

- Introduction
- Executive Summary
- Metadata
- Data Wrangling
- Data Transformation
- In-Depth Examination
- Data Visualization
- Insights
- Considerations
- Recommendations

Introduction

- Bull and Bear markets reflect the overall direction and health of the stock market.
- Understanding these trends helps investors decide when to buy, hold, or sell stocks.
- They signal broader economic conditions—bull markets are linked to economic growth and high confidence, while bear markets often indicate downturns and reduced investor sentiment.
- Businesses use these indicators to guide financing, expansion decisions, and strategic planning. Policy Formulation: Economists and policymakers monitor these trends to shape monetary and fiscal policies accordingly.

Executive Summary

Project Objective:

The objective of this project is to create a dashboard showing financial metric, employee metric and stock information for major companies as well as analyze bull and bear markets to get insights on the overall health and direction of the stock market

Key Insights:

- NVIDIA achieved an exceptional CAGR of 77.57% over 10 years, showcasing outstanding long-term growth.
- META and Eli Lilly lead their respective exchanges (NASDAQ and NYSE) in stock price, reflecting strong investor confidence.
- Apple, JPMorgan, and Tesla show consistent growth in Free Cash Flow, indicating solid operational efficiency.

Immediate Future Actions:

- Invest in high-growth and financially stable companies such as NVIDIA, Apple, and JPMorgan, while monitoring recovery trends in firms that dipped in 2022 like Microsoft and Google.
- Enhance operational efficiency by benchmarking sales per employee (e.g., Meta), managing workforce size strategically (e.g., Amazon, Tesla), and maintaining a balanced capital structure.
- Align executive pay and workforce strategy with long-term business goals by linking compensation to performance and continuing to build talent in growth-driven companies like Tesla and Apple.

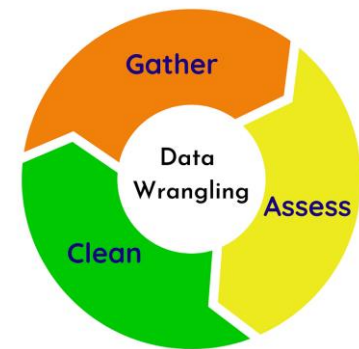
About the Dataset

- The open-source dataset is taken from Kaggle.
- This dataset offers the financial stock information of major company contained in three tables
- The profile table shares the geographical, financial and stock related information for the tickers
- The price history table gives the dates and prices of ticker
- The stock financials includes the cash, debt, revenue, profit margin, net income etc. information of the tickers.

Data Wrangling

The csv data file was extracted with Power BI. The Data Wrangling operations were performed using Power Query. Some of the performed operations were:

- Header Promotion
- Filtering Columns
- Correcting Data Types
- Expanding Columns



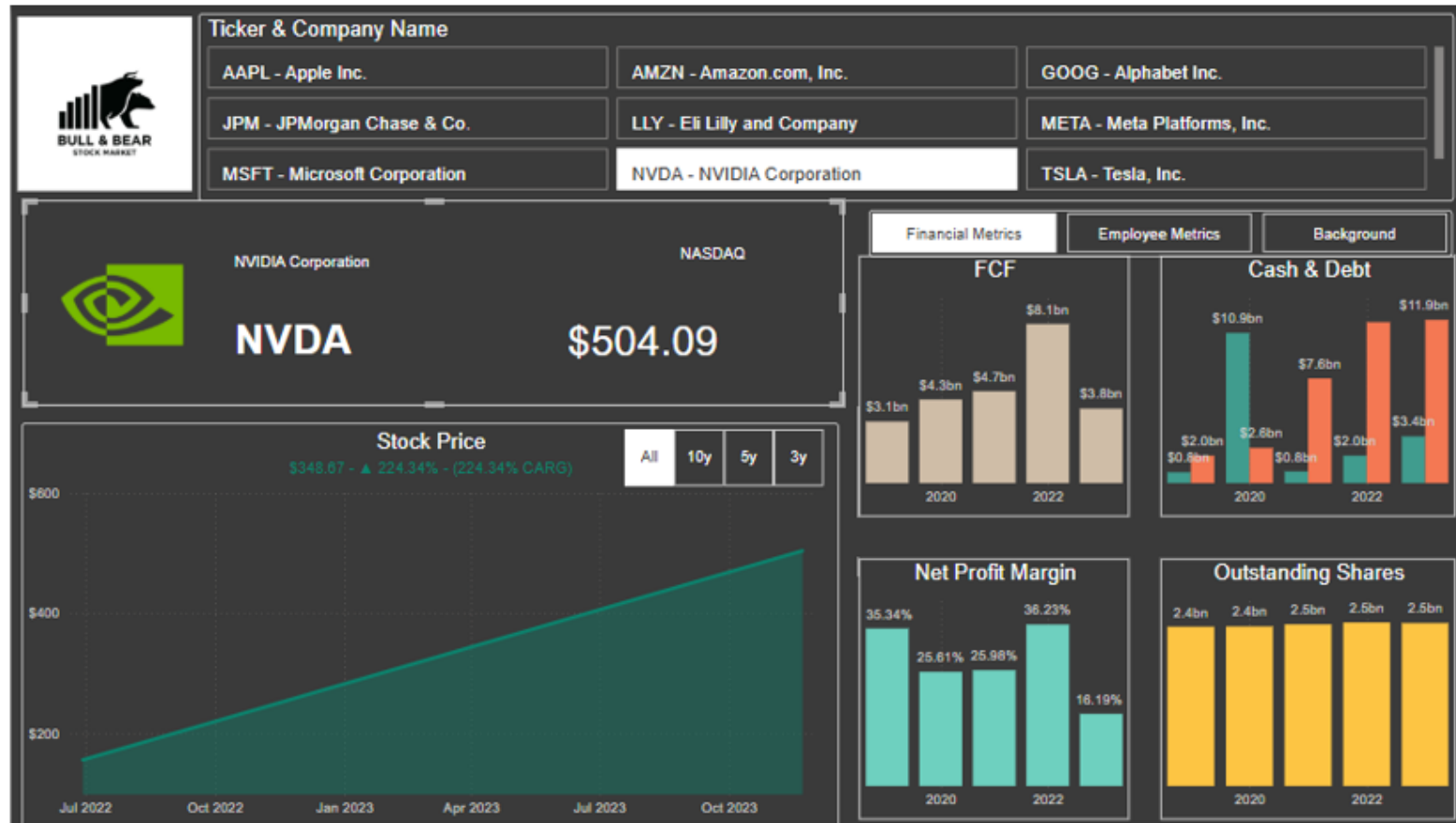
Data Visualization



Dashboard Breakdown

National
Association of
Securities
Dealers

Stock price
data over
time

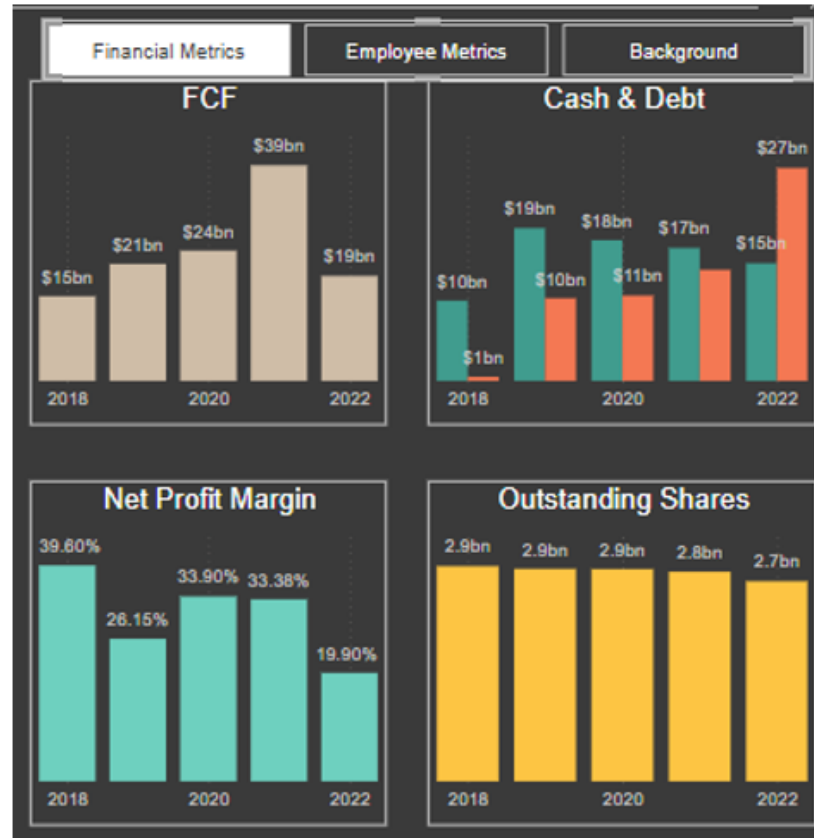


Filtering
from
companies

Financial,
Employee
and
Company
Information

Financial Metrics

Free Cash Flow refers to the amount of cash a company generates after accounting for capital expenditures



The percentage of net income (profit) a company earns from its total revenue.

Cash and Debt compares the cash being earned and debt owed

The total number of a company's shares that are currently held by all shareholders

Employee Metrics

Employee Count

66,185 Current Employees

Sales to Employee
issue

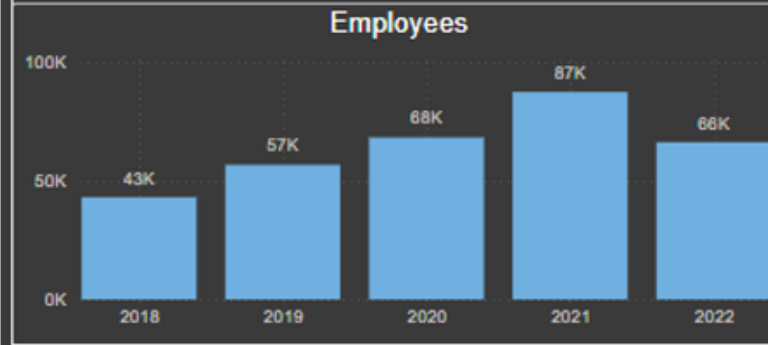
\$1.78M
Sales/Employee

-21K
Employees Added

27M
CEO Compensat...

CEO
Compensation

Number of Employees
per year



Background

Company Information



The screenshot shows a web application interface with three tabs: 'Financial Metrics', 'Employee Metrics', and 'Background'. The 'Background' tab is selected and highlighted. Below the tabs, the text describes Nvidia Corporation as an American multinational corporation headquartered in Santa Clara, California, and incorporated in Delaware. It also mentions the company's founding in 1993 by Jensen Huang, Chris Malachowsky, and Curtis Priem, and its focus on designing and supplying graphics processing units (GPUs), application programming interfaces (APIs), and system on a chip units (SoCs) for mobile computing and the automotive market. Finally, it states that Nvidia is a leading supplier of artificial intelligence (AI) hardware and software, outsourcing the manufacturing of the hardware it designs.

Financial Metrics Employee Metrics **Background**

Nvidia Corporation is an American multinational corporation and technology company headquartered in Santa Clara, California, and incorporated in Delaware.

Founded in 1993 by Jensen Huang (president and CEO), Chris Malachowsky, and Curtis Priem, it designs and supplies graphics processing units (GPUs), application programming interfaces (APIs) for data science and high-performance computing, and system on a chip units (SoCs) for mobile computing and the automotive market.

The company is also a leading supplier of artificial intelligence (AI) hardware and software. Nvidia outsources the manufacturing of the hardware it designs.

Analysis Insights



- 1) META recorded the highest stock price on the NASDAQ, trading at \$339.97, reflecting strong investor confidence in its market positioning and growth potential.
- 2) Eli Lilly (LLY) led the NYSE listings with the highest stock price at \$597.60, highlighting its strong performance in the pharmaceutical sector.
- 3) NVIDIA demonstrated the highest Compound Annual Growth Rate (CAGR) over the past 10 years, achieving an impressive 77.57%, indicating exceptional long-term growth.
- 4) Apple, JPMorgan, and Tesla have shown a consistent increase in Free Cash Flow (FCF) over the years, suggesting strong operational efficiency and cash generation.

Analysis Insights



- 5) Microsoft, Meta, Nvidia, and Google experienced a notable dip in financial performance in 2022, following a previously consistent upward trend.
- 6) JPMorgan and Tesla stand out as the only companies with higher cash reserves than debt, demonstrating robust balance sheet strength and financial stability.
- 7) While Apple, Eli Lilly, and Tesla exhibit a rising net profit margin, Microsoft maintains the highest net profit margin overall, signaling superior profitability.
- 8) Tesla has seen a steady increase in outstanding shares, whereas other companies have either maintained or reduced theirs. Apple holds the largest number of outstanding shares in absolute terms.

Analysis Insights



- 9) Amazon employs the largest workforce among the companies analyzed, underscoring its operational scale.
- 10) Meta leads in sales per employee, indicating high workforce productivity and efficient revenue generation.
- 11) Tesla added the most employees, while Amazon underwent the largest workforce reduction, suggesting divergent strategic directions in workforce management.
- 12) In terms of executive compensation, Google's CEO received the highest remuneration, followed by Apple and Microsoft, reflecting their leadership valuation.
- 13) Tesla and Apple have demonstrated a steady upward trend in employee growth, reinforcing their long-term expansion strategies.

Considerations to keep in mind

- 1) The dataset is not inclusive of all the details regarding all the companies include. As a result, the accuracy of this information is confined to the scope of the provided dataset.
- 2) Apart from the discussed metrics there are many other metrics that effect the bull and bear stock company.



Recommendations



- 1) Prioritize NVIDIA, Tesla, and Meta for long-term growth investment based on strong CAGR and stock performance.
- 2) Consider Apple, Microsoft, and JPMorgan for stable returns due to strong free cash flow and profitability.
- 3) Exercise caution with Microsoft, Meta, Nvidia, and Google, which showed a dip in 2022 after prior consistent growth.
- 4) Companies like Amazon should optimize workforce efficiency and review productivity due to high employee count and recent reductions.
- 5) Meta's high sales per employee serves as a benchmark for improving organizational efficiency.

Recommendations



- 6) Tesla should manage its increasing outstanding shares to prevent shareholder dilution.
- 7) Companies with stable or declining outstanding shares may continue leveraging share buyback strategies.
- 8) JPMorgan and Tesla should capitalize on strong cash positions and low debt to invest in growth, innovation, or shareholder returns.
- 9) Align CEO compensation with long-term value creation, especially for firms like Google, Apple, and Microsoft.
- 10) Tesla and Apple should maintain their momentum in talent acquisition while ensuring it aligns with long-term strategic growth.
- 11) Companies reducing workforce should focus on upskilling, automation, and retaining high-performing employees.

THE END
